

Chapter 1

Sequences

A sequence is one of the most fundamental objects in mathematical analysis. Informally, a sequence is an ordered list of numbers

$$a_1, a_2, a_3, \dots$$

in which the position of each number is determined by a positive integer. Sequences are particularly useful for studying approximation and the notion of convergence.

1.0.1 Definition of a Sequence

Definition 1.1 (Sequence). A sequence of real numbers *is a function*

$$a : \mathbb{N} \longrightarrow \mathbb{R}.$$

We usually denote the sequence by

$$(a_n)_{n \in \mathbb{N}} \quad \text{or simply} \quad (a_n)_{n \geq 1},$$

where

$$a_n = a(n)$$

is called the n -th term of the sequence.

Example 1.2 (Arithmetic/geometric/harmonic sequence). An arithmetic sequence is a sequence in which the difference between two consecutive terms is constant. Thus,

$$a_{n+1} - a_n = d$$

for some constant $d \in \mathbb{R}$. The general form of an arithmetic sequence is

$$a_n = a_1 + (n - 1)d.$$

A geometric sequence is a sequence in which the ratio between consecutive terms is constant. Thus,

$$\frac{a_{n+1}}{a_n} = r$$

whenever $a_n \neq 0$, where $r \in \mathbb{R}$ is called the common ratio. The general form is

$$a_n = a_1 r^{n-1}.$$

The harmonic sequence is the sequence

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots,$$

whose n -th term is

$$a_n = \frac{1}{n}.$$

A sequence can also be defined recursively. For example,

$$a_1 = 1, \quad a_{n+1} = \frac{a_n + 2}{2}$$

defines a sequence by specifying its first term and then describing how each subsequent term is obtained from the preceding one.

Example 1.3 (Fibonacci sequence). The Fibonacci sequence is defined recursively by

$$F_1 = 1, \quad F_2 = 1,$$

and

$$F_{n+2} = F_{n+1} + F_n \quad \text{for } n \geq 1.$$

Thus,

$$1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$$

are the first few Fibonacci numbers.

1.0.2 Increasing and Decreasing Sequences

The way in which the terms of a sequence change as n increases is often important.

Definition 1.4 (Increasing sequence). A sequence (a_n) is called increasing if

$$a_n \leq a_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

If

$$a_n < a_{n+1} \quad \text{for every } n \in \mathbb{N},$$

then (a_n) is called strictly increasing.

Similarly,

Definition 1.5 (Decreasing sequence). A sequence (a_n) is called decreasing if

$$a_n \geq a_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

If

$$a_n > a_{n+1} \quad \text{for every } n \in \mathbb{N},$$

then (a_n) is called strictly decreasing.

A sequence which is either increasing or decreasing is called a *monotone sequence*.

For example, the sequence

$$a_n = 1 - \frac{1}{n}$$

is strictly increasing, since

$$1 - \frac{1}{n} < 1 - \frac{1}{n+1}.$$

On the other hand,

$$b_n = \frac{1}{n}$$

is strictly decreasing.

1.0.3 Bounded Sequences

Another important property of a sequence is whether its terms remain within some fixed range.

Definition 1.6 (Bounded above). *A sequence (a_n) is said to be bounded above if there exists a number $M \in \mathbb{R}$ such that*

$$a_n \leq M \quad \text{for every } n \in \mathbb{N}.$$

The number M is called an upper bound of the sequence.

Similarly,

Definition 1.7 (Bounded below). *A sequence (a_n) is said to be bounded below if there exists a number $m \in \mathbb{R}$ such that*

$$a_n \geq m \quad \text{for every } n \in \mathbb{N}.$$

The number m is called a lower bound of the sequence.

Definition 1.8 (Bounded sequence). *A sequence (a_n) is called bounded if it is bounded both above and below. Equivalently, there exists a constant $C > 0$ such that*

$$|a_n| \leq C \quad \text{for every } n \in \mathbb{N}.$$

For example,

$$a_n = \frac{1}{n}$$

is bounded, since

$$0 < a_n \leq 1$$

for every n . In contrast, the sequence

$$a_n = n$$

is not bounded above.

1.0.4 Limits of Sequences

One of the central questions concerning a sequence is whether its terms approach a particular number as n becomes large.

Definition 1.9 (Limit of a sequence). *Let (a_n) be a sequence of real numbers. We say that (a_n) converges to $L \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|a_n - L| < \varepsilon \quad \text{for every } n \geq N.$$

In this case, we write

$$\lim_{n \rightarrow \infty} a_n = L$$

or

$$a_n \longrightarrow L \quad \text{as } n \rightarrow \infty.$$

The definition says that the terms of the sequence can be made arbitrarily close to L by taking n sufficiently large.

For example,

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Indeed, given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then, for every $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

A sequence which has a finite limit is called a *convergent sequence*. If a sequence does not converge to any real number, it is called *divergent*.

An important consequence of the definition is that a convergent sequence is always bounded.

Theorem 1.10. *Every convergent sequence of real numbers is bounded.*

Proof. Suppose that

$$a_n \longrightarrow L.$$

Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$|a_n - L| < 1 \quad \text{for all } n \geq N.$$

Hence

$$|a_n| \leq |a_n - L| + |L| < 1 + |L| \quad \text{for all } n \geq N.$$

The finitely many terms

$$a_1, a_2, \dots, a_{N-1}$$

are also bounded. Therefore, the entire sequence (a_n) is bounded. $\square$

Theorem 1.11 (Uniqueness of limits). *If a sequence (a_n) converges, then its limit is unique. In other words, if*

$$a_n \longrightarrow L \quad \text{and} \quad a_n \longrightarrow M,$$

then

$$L = M.$$

Proof. Suppose, for contradiction, that $L \neq M$. Set

$$\varepsilon = \frac{|L - M|}{3} > 0.$$

Since $a_n \rightarrow L$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - L| < \varepsilon \quad \text{for all } n \geq N_1.$$

Similarly, since $a_n \rightarrow M$, there exists $N_2 \in \mathbb{N}$ such that

$$|a_n - M| < \varepsilon \quad \text{for all } n \geq N_2.$$

Therefore, for every $n \geq \max\{N_1, N_2\}$, the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < 2\varepsilon = \frac{2}{3}|L - M|.$$

This is impossible. Hence

$$L = M.$$

$\square$

Limits of sequences satisfy several useful algebraic rules. These rules allow us to compute the limit of a complicated sequence from the limits of simpler sequences.

Laws of limits

Theorem 1.12 (Algebraic laws of limits). *Suppose that*

$$a_n \longrightarrow a \quad \text{and} \quad b_n \longrightarrow b$$

as $n \rightarrow \infty$. Then:

1. Addition:

$$a_n + b_n \longrightarrow a + b.$$

2. Subtraction:

$$a_n - b_n \longrightarrow a - b.$$

3. Multiplication:

$$a_n b_n \longrightarrow ab.$$

4. Multiplication by a constant: *For any $c \in \mathbb{R}$,*

$$ca_n \longrightarrow ca.$$

5. Division: *If $b \neq 0$, then*

$$\frac{a_n}{b_n} \longrightarrow \frac{a}{b},$$

provided $b_n \neq 0$ for all sufficiently large n .

6. Absolute value:

$$|a_n| \longrightarrow |a|.$$

7. Exponentiation: *For a positive $k > 0$,*

$$\lim_{n \rightarrow \infty} a_n^k = \left(\lim_{n \rightarrow \infty} a_n \right)^k.$$

Theorem 1.13 (Squeeze theorem). *Suppose that (a_n) , (b_n) and (c_n) are sequences such that*

$$a_n \leq b_n \leq c_n$$

for all sufficiently large n . If

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L,$$

then

$$\lim_{n \rightarrow \infty} b_n = L.$$

The squeeze theorem is particularly useful when the middle sequence is difficult to handle directly. For example,

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n},$$

and since

$$\lim_{n \rightarrow \infty} -\frac{1}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0,$$

the squeeze theorem gives

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0.$$

Some Important Limits

The following limits are fundamental and will be used repeatedly.

Theorem 1.14 (Exponential versus polynomial growth). *Let $a > 1$ and $p > 0$. Then*

$$\lim_{n \rightarrow \infty} \frac{n^p}{a^n} = 0.$$

Equivalently,

$$\lim_{n \rightarrow \infty} \frac{a^n}{n^p} = +\infty.$$

Thus, an exponential sequence with base greater than 1 grows faster than any positive power of n .

For example,

$$\lim_{n \rightarrow \infty} \frac{n^3}{2^n} = 0, \quad \lim_{n \rightarrow \infty} \frac{n^{100}}{10^n} = 0.$$

Theorem 1.15 (Two important limits). *We have*

$$\lim_{k \rightarrow \infty} k^{1/k} = 1.$$

The sequence $\left(1 + \frac{1}{k}\right)^k$ converges. Its limit is denoted by e :

$$e = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k.$$

The number e is approximately

$$e \approx 2.718281828.$$

Divergent Sequences

A sequence is called divergent if it does not converge to any finite real number.

Definition 1.16 (Divergent sequence). *A sequence (a_n) is called divergent if there is no $L \in \mathbb{R}$ such that*

$$a_n \longrightarrow L.$$

There are several different ways in which a sequence can diverge. For example, a sequence may become arbitrarily large, arbitrarily negative, or oscillate without approaching a single number.

Definition 1.17 (Divergence to $+\infty$). *We say that*

$$a_n \longrightarrow +\infty$$

if, for every $M > 0$, there exists $N \in \mathbb{N}$ such that

$$a_n > M \quad \text{for all } n \geq N.$$

Similarly,

Definition 1.18 (Divergence to $-\infty$). *We say that*

$$a_n \longrightarrow -\infty$$

if, for every $M < 0$, there exists $N \in \mathbb{N}$ such that

$$a_n < M \quad \text{for all } n \geq N.$$

For example,

$$n^2 \longrightarrow +\infty, \quad -n^3 \longrightarrow -\infty.$$

However, divergence does not necessarily mean that the terms tend to $+\infty$ or $-\infty$. For example,

$$a_n = (-1)^n$$

is divergent because it oscillates between -1 and 1 .

Remark 1.19. *A sequence can therefore be classified into three broad types:*

1. *it converges to a finite real number L ;*
2. *it diverges to $+\infty$ or $-\infty$;*
3. *it is divergent without tending to either $+\infty$ or $-\infty$, for example because it oscillates.*

1.0.5 Subsequences

Sometimes we are interested not in every term of a sequence, but only in certain terms. This leads to the notion of a subsequence.

Definition 1.20 (Subsequence). *Let (a_n) be a sequence. A subsequence of (a_n) is a sequence of the form*

$$(a_{n_k})_{k \geq 1},$$

where

$$n_1 < n_2 < n_3 < \dots$$

is a strictly increasing sequence of positive integers.

Thus, a subsequence is obtained by selecting infinitely many terms from the original sequence while preserving their original order.

For example, if

$$a_n = n,$$

then

$$a_{2k} = 2k, \quad k = 1, 2, 3, \dots,$$

is a subsequence consisting of the even-indexed terms:

$$2, 4, 6, 8, \dots$$

Similarly,

$$a_{2k-1} = 2k - 1$$

is the subsequence of odd-indexed terms.

A useful fact about convergent sequences is that every subsequence has the same limit.

Theorem 1.21. *If*

$$a_n \longrightarrow L,$$

then every subsequence (a_{n_k}) also converges to L .

Theorem 1.22 (Subsequence criterion for convergence). *A sequence (a_n) converges to L if and only if every subsequence (a_{n_k}) has a further subsequence that converges to L .*

Proof. We prove the two implications separately.

($\Rightarrow$) Suppose that

$$a_n \longrightarrow L.$$

Let (a_{n_k}) be any subsequence of (a_n) . We know that every subsequence of a convergent sequence converges to the same limit. Therefore,

$$a_{n_k} \longrightarrow L.$$

In particular, (a_{n_k}) itself is a further subsequence of (a_{n_k}) that converges to L .

($\Leftarrow$) Conversely, suppose that every subsequence (a_{n_k}) has a further subsequence that converges to L . We prove that

$$a_n \longrightarrow L$$

by contradiction. Suppose that (a_n) does not converge to L . Then, by the negation of the definition of convergence, there exists $\varepsilon_0 > 0$ such that for every $N \in \mathbb{N}$, there exists $n \geq N$ satisfying

$$|a_n - L| \geq \varepsilon_0.$$

We can use this property to construct a subsequence (a_{n_k}) such that

$$|a_{n_k} - L| \geq \varepsilon_0 \quad \text{for every } k \in \mathbb{N}.$$

Indeed, choose n_1 such that

$$|a_{n_1} - L| \geq \varepsilon_0.$$

Having chosen n_k , use the above property with $N = n_k + 1$ to choose $n_{k+1} > n_k$ such that

$$|a_{n_{k+1}} - L| \geq \varepsilon_0.$$

By our assumption, the subsequence (a_{n_k}) has a further subsequence $(a_{n_{k_j}})$ which converges to L . Hence, by the definition of convergence, there exists $J \in \mathbb{N}$ such that

$$|a_{n_{k_j}} - L| < \varepsilon_0 \quad \text{for all } j \geq J.$$

But this contradicts the construction of the subsequence, for which

$$|a_{n_{k_j}} - L| \geq \varepsilon_0 \quad \text{for every } j.$$

Therefore, our assumption was false, and hence

$$a_n \longrightarrow L.$$

□

1.0.6 The Bolzano–Weierstrass Theorem

The following fundamental theorem connects boundedness with the existence of convergent subsequences.

Theorem 1.23 (Bolzano–Weierstrass). *Every bounded sequence of real numbers has a convergent subsequence.*

In other words, if (a_n) is a sequence for which there exists $C > 0$ such that

$$|a_n| \leq C \quad \text{for every } n \in \mathbb{N},$$

then there exist a subsequence (a_{n_k}) and a number $L \in \mathbb{R}$ such that

$$a_{n_k} \longrightarrow L \quad \text{as } k \rightarrow \infty.$$

It is important to note that the theorem does *not* say that every bounded sequence converges. For example, the sequence

$$a_n = (-1)^n$$

is bounded, since

$$|a_n| = 1$$

for every n , but it does not converge. However, it has convergent subsequences:

$$a_{2k} = 1 \longrightarrow 1, \quad a_{2k-1} = -1 \longrightarrow -1.$$

Thus, the Bolzano–Weierstrass theorem guarantees that boundedness is enough to extract at least one convergent subsequence, even when the original sequence itself does not converge.

Summary

The main concepts introduced in this section can be summarized as follows:

- A *sequence* is a function $a : \mathbb{N} \rightarrow \mathbb{R}$.
- A sequence is *increasing* if $a_n \leq a_{n+1}$ and *decreasing* if $a_n \geq a_{n+1}$.
- A sequence is *bounded* if there exists $C > 0$ such that $|a_n| \leq C$ for every n .
- A sequence (a_n) *converges* to L if, for every $\varepsilon > 0$, all sufficiently large terms satisfy $|a_n - L| < \varepsilon$.
- A *subsequence* is obtained by selecting terms $a_{n_1}, a_{n_2}, \dots$ with $n_1 < n_2 < \dots$.
- Every convergent sequence is bounded.
- Every subsequence of a convergent sequence converges to the same limit.
- By the *Bolzano–Weierstrass theorem*, every bounded sequence of real numbers has a convergent subsequence.